

Surveyor App PRD

Problem statement:

Property owners and other customers struggle to discover and book qualified land surveyors because the market is fragmented across word-of-mouth referrals, local contacts, and informal communication channels. This lack of centralized discoverability makes it difficult to find the right surveyor, confirm availability, and schedule services efficiently, while limiting surveyors' ability to reach customers beyond their existing local networks.

Goals & NonGoals:

Goals: Compare and book the services of Expert Landsurveyors registered on platform, which lists year of experience, tools or methods implemented for digital survey, location and availability on selected date.

Post Survey reports will be delivered to customer via platform in downloadable Pdf files format. Interrupted survey meetings due to bad weather, insufficient time or bad light, other unforeseen blockages needs to be allotted to same Expert until completed.

NonGoals:

Price negotiation with Expert, Dealing issues with neighbouring owners incase of disputes, Verification /validation of land ownership is not in scope.

Success metric :

NorthStar - Number of completed bookings by users.

L1 : % of bookings accepted by expert, % of bookings received final pdf files.

Counter Metric: Task drop rate, D30 inactivation rate

UserPersona :

CASE STUDY — USER PERSONA



User Persona 1: Primary

Name (label: "User") + short 1-line life with the product ("A day in one sentence")
Rohan Sharma | Needs clear property boundaries to build a new boundary wall safely.

Demographics

- Age Range: 35-45 years
- Location: Bengaluru Suburbs
- Occupation: IT Project Manager
- Income Band: ₹15-25 LPA
- Relationship/Household: Married, 2 children
- Digital Literacy: Good (uses apps, online banking)

Psychographic

- Pains (🔴): Unclear pricing, unlicensed 'experts', risk of boundary disputes.
- Gains (🟢): Confident legal documentation, verified surveyor ratings, transparent booking process.
- Motivations: Secure property investment, avoid legal issues.
- Values: Legality, professional quality, efficiency. Fears: Scams, hidden costs, future legal trouble.

Tech Proficiency

- Devices: iPhone, Windows Laptop
- OS: iOS, Windows 11
- App Habits: Uses apps for banking, travel, food delivery.
- Social Media: Active on LinkedIn, Instagram.
- Internet Constraints: Moderate data plan, fast home Wi-Fi.
- Accessibility Needs: None specified.

Product-Specific

- Current Solutions: Local directories, Google searches, word-of-mouth references.
- Usage Scenarios & Frequency: Once every few years (property purchase, renovation).
- Purchase Drivers: Trust, verified license, clear availability.
- Satisfaction Level: Low with existing, unorganized methods.
- Switching Barriers: Lack of time to research, difficult verification.
- Success Metric: Valid, legal survey map delivered quickly.

What they want

- Transparent, all-inclusive pricing.
- Verified & licensed local experts.
- Book on mobile with confirmed slots.
- Fast, digital delivery of survey maps.

User Persona 2: Secondary

Name (label: "User") + short 1-line life with the product ("A day in one sentence")
Anjali Gupta | Rapidly verifies multiple land plots during client site-planning meetings.

Demographics

- Age Range: 40-50 years
- Location: Mumbai Metropolitan Region
- Occupation: Real Estate Developer
- Income Band: ₹40-60 LPA (Business Owner)
- Relationship/Household: Single-person business owner
- Digital Literacy: High (uses business tools, project apps)

Psychographic

- Pains (🔴): Slow surveyor response times, incomplete digital maps, difficult urgent bookings.
- Gains (🟢): Immediate availability check, direct digital data download, priority corporate service.
- Motivations: Expand development portfolio, fast client project delivery.
- Values: Reliability, speed, professional data.
- Fears: Missing construction deadlines, expensive planning errors.

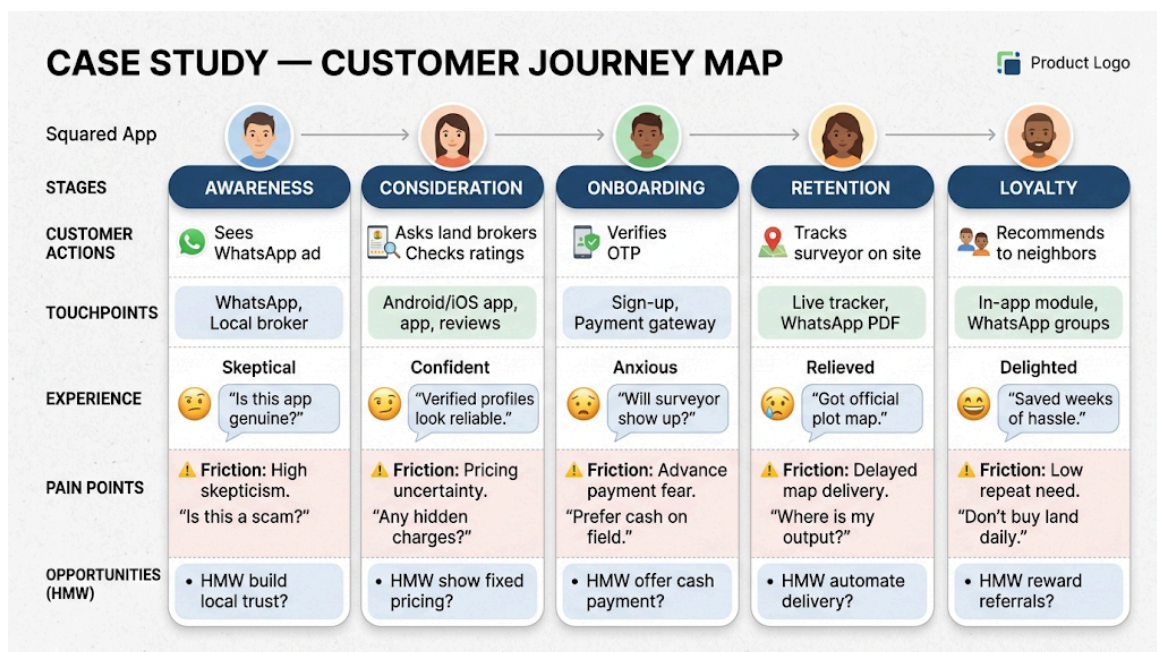
Product-Specific

- Current Solutions: Maintains a small list of freelance surveyors.
- Usage Scenarios & Frequency: Multiple times per month (different projects).
- Purchase Drivers: Speed, reliable data quality, digital workflows.
- Satisfaction Level: Variable (depends on individual surveyor).
- Switching Barriers: Existing relationships, high cost of wrong decisions.
- Success Metric: Quick turnaround from booking to data delivery.

What they want

- Fast-track booking options for urgent jobs.
- Direct download of survey data maps.
- Bulk or corporate service accounts.
- Clear service level agreements (SLAs).

Userjourney:



Competitor Research :

	Property Record Platforms	Service Marketplaces	Legal / Due Diligence	Squared
Discover Surveyors	✗	●	● Referral	✓ Direct
Compare & Select	✗	●	✗	✓
Check Availability	✗	✓	✗	✓ Real-time
Book Survey	✗	●	● Manual	✓
Survey Tracking	✗	✓	✗	✓
Digital Survey Report	● Records	✗	● Manual	✓ CAD / PDF
Land Records	✓ Core	✗	●	● Supporting
Legal / Title Review	●	✗	✓ Core	● Partner-led

Feature / Initiative	Reach(Users / Mo)	Impact(0.25 - 3.0)	Confidence(50% - 100%)	Effort(Person-Weeks)	Priority Rank
Pin-Drop & FMB Boundary Map Overlay	4000	2.0	80	2	P2
Government-Verified Surveyor Badges	5000	3.0	80	4	P0
Upfront Package	3000	2.0	50	4	P1

Pricing & Escrow Payment					
B2B Developer Subscription Accounts	1000	1.0	100	4	P3

- 3.0 = Massive impact (Primary conversion driver) 2.0 = High impact 1.0 = Medium impact 0.5 = Low impact 0.25 = Minimal impact

Functional & Non Functional Requirements

Dual language Telugu and English usability. Pdf ,cad files downloadable with tamperproof watermarking. Service for partitions in joint family land parcel surveying. Fair pricing calculated with given map coordinates. Complex village names can be entered by voice option.

NonFunctional: PQRSTU framework

Performance: within 600 ms time to load the available surveyors for selected region and date time slot. #Quality : Government certified expertise validation after checking certificates and licenses. #Reliability: All the data will be encrypted and stored only for 6 months. # Security: DPDP act implementation for secure data and transactions in escrow account method. #Throughput: #Useability: User should search surveyor expert and book within 4 minutes;

Acceptance criteria

- **AC 1.1:** Display the "Licensed Revenue Surveyor" badge on profile cards if and only if `is_verified == true` in the admin database.
- **AC 1.2:** Tapping the verification badge opens a modal displaying the license ID, issuing authority, and expiration date.
- **AC 2.1: Given** a user enters a valid Telangana (*Dharani*) or AP (*Meebhoomi*) survey/Pahani number, **When** submitted, **Then** the official Field Measurement Book (FMB) boundary polygon overlays onto the satellite map in **< 2 seconds**.
- **AC 2.2: Given** a user drops a GPS pin on the satellite view, **When** confirmed, **Then** auto-populate the nearest village name, mandal, district, and survey number within a 50-meter radius.
- **AC 3.1:** Auto-calculate package cost at checkout based on selected service type (*Boundary Demarcation, Sub-division, Topography*) and plot size (acres/sq. yards).
- **AC 3.2:** When selecting "Escrow / Pay-After-Survey", authorize the transaction amount without transferring funds to the surveyor balance until job completion.
- **AC 4.1:** Upon booking confirmation, send instant SMS and WhatsApp notifications containing slot details, surveyor contact, and booking ID to both user and surveyor.
- **AC 5.1:** Upon CAD/PDF file upload by the surveyor, trigger a secure download link delivered to the user's registered WhatsApp number within **< 5 seconds**.
- **AC 5.2:** Watermark draft preview maps with "UNAPPROVED DRAFT" across all pages until the client signs off or escrow is cleared.

MockUp

Prototype

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